Self-Supervised Learning for Chilli Leaf Disease Detection: A Comparative Study with DeiT-Small Backbone

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***Abstract*—Agricultural crop disease leads to significant yield losses globally, and automated plant disease classification from images continues to be difficult in low-label environments. This paper examines two advanced self-supervised learning (SSL) frameworks—Bootstrap Your Own Latent (BYOL) and Self-Distillation with No Labels (DINO)—utilised on the Chilli Disease 2024 dataset, which consists of 10,987 images categorised into five classes. Both methods utilise a DeiT-Small Vision Transformer backbone pre-trained in a completely unsupervised manner on 80% of the data. DINO achieves a linear probe accuracy of 88.81% and a macro F1 of 0.8878. BYOL achieves an accuracy of 80.89% and a macro F1 of 0.8098 with the same evaluation method. k-NN tests show that DINO produces embeddings with superior geometric structure (k=1 accuracy: 95.27%) compared to BYOL (72.07%). Qualitative analysis of DINO's self-attention maps demonstrates semantically significant localisation of disease regions devoid of label supervision. These findings illustrate that self-supervised pre-training utilising Vision Transformers can achieve performance comparable to supervised methods while significantly decreasing annotation demands.**

***Index Terms*—Self-supervised learning, BYOL, DINO, Vision Transformer, DeiT-Small, plant disease detection, chilli disease, linear probing, k-NN evaluation**

I. Introduction

Diseases that affect crops are one of the main reasons why crops fail around the world. It is thought that pests and pathogens cause 20–40% of annual crop yields to be lost. Chilli (*Capsicum* spp.) is a high-value spice crop grown all over Asia and Latin America, vulnerable to a number of foliar diseases such as cercospora leaf spot, powdery mildew, mite and thrips infestations, and nutritional deficiencies. It is very important to find these conditions quickly so that the right treatment can be given, but visual scouting in the field is hard work and can lead to mistakes. Deep learning and computer vision have shown a lot of promise for automating the diagnosis of plant diseases. However, the most accurate supervised methods need large datasets with annotations.

The main question this paper addresses is: how can we train useful visual representations for classifying chilli diseases when there is limited labelled data? Self-supervised learning (SSL) provides an effective approach by utilising unannotated raw visual structure. BYOL [1] acquires knowledge through bootstrapping predictions between an online network and an EMA target network, circumventing mode collapse without negative pairs. DINO [2] utilises a teacher-student knowledge distillation framework with multi-crop augmentation and a high-dimensional prototypical head, yielding representations with emergent semantic segmentation characteristics.

Prior work on plant disease detection has predominantly relied on supervised CNNs. The application of SSL methods, and particularly Vision Transformers, to agricultural disease datasets remains underexplored. This study contributes: (i) a rigorous comparison of BYOL and DINO using identical DeiT-Small backbones on the Chilli Disease 2024 dataset; (ii) a label-fraction ablation study; (iii) qualitative DINO attention map analysis for disease localisation; and (iv) comparison against fully supervised baselines. We hypothesise that DINO's multi-crop strategy and ViT-specific design will yield geometrically superior embeddings.

This paper is organised as follows: Section II reviews related literature. Section III describes the methodology. Section IV presents results. Section V discusses findings. Section VI concludes with future directions.

II. Literature Review

Grill et al. [1] proposed BYOL, a self-supervised method that trains an online network to predict the output of an EMA target network from a different augmented view. BYOL requires no negative pairs and achieves 74.3% top-1 accuracy on ImageNet with ResNet-50, outperforming SimCLR. A known limitation is sensitivity to batch normalisation and learning rate schedules, which can cause instability on smaller datasets.

Caron et al. [2] introduced DINO, a self-distillation framework for ViTs using cross-entropy loss with centring and temperature sharpening. Multi-crop augmentation is a central design element. DINO features exhibit emergent segmentation through attention maps and achieve 80.1% linear probe accuracy on ImageNet with ViT-S/16. A known drawback is the additional memory cost of multi-crop training.

Oquab et al. [3] introduced DINOv2, enhancing DINO with carefully curated large-scale data, KoLeo regularisation, and knowledge distillation from larger teacher models. DINOv2 ViT-S/14 achieves 81.1% on ImageNet linear probing, but requires substantially greater computational resources, limiting its utility in low-resource domain-specific settings.

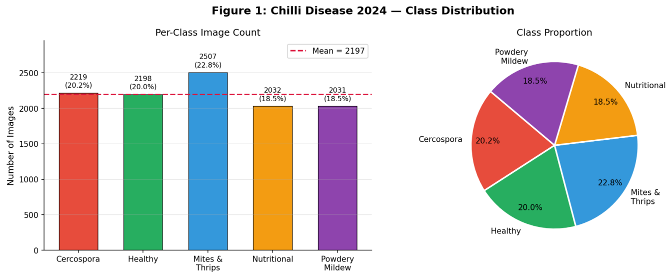

Fig. 1. Chilli Disease 2024 class distribution: bar chart (left) and proportion pie chart (right).

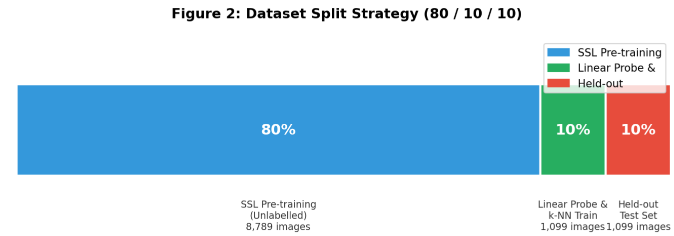

Fig. 2. Dataset split strategy. SSL pre-training uses 80% unlabelled data; downstream evaluation uses 10% labelled probe and 10% held-out test sets.

Chen et al. [4] proposed SimCLR, utilising contrastive learning with a projection head to attain 76.5% top-1 accuracy with ResNet-50 on ImageNet. The main limitation is a requirement for very large batches (4096+), which is impractical for domain-specific datasets.

He et al. [5] proposed MAE, a reconstruction-based self-supervised method in which a ViT encoder is trained to reconstruct masked image patches, achieving 83.1% linear probe accuracy with ViT-B on ImageNet, without providing DINO-style attention visualisations.

Dosovitskiy et al. [6] introduced the Vision Transformer (ViT), applying the transformer architecture to image patches. ViT-L/16 achieves 87.1% on ImageNet when pre-trained on JFT-300M, motivating SSL strategies to compensate for limited labelled data.

Tran et al. [7] compiled the Chilli Disease 2024 dataset comprising 10,987 images across five categories. The original study employed standard CNN classifiers; no prior research has applied SSL techniques to this dataset, representing a significant research gap addressed in the present work.

Misra and van der Maaten [8] proposed PIRL, a contrastive objective-based method for learning context-invariant representations, achieving 63.6% top-1 accuracy with ResNet-50. PIRL requires a memory bank and does not utilise ViT architectures, making it less efficient than EMA-based methods.

III. Methodology

A. Dataset and Initial Data Analysis

The Chilli Disease 2024 dataset [7] contains 10,987 RGB images across five classes (Mendeley Data, DOI: 10.17632/tf9dtfz9m6.2). The class distribution is approximately balanced with an imbalance ratio of 1.23. Mean R, G, and B channel analysis of 300 sampled images confirms a green-dominant colour profile characteristic of plant imagery. No duplicates or missing values were found. The class distribution is shown in Fig. 1.

B. Data Splitting

Stratified splitting (seed = 42) was used to preserve class proportions across all partitions. The SSL pre-training pool comprised 8,789 images with labels discarded; the linear probe and k-NN training set contained 1,099 labelled images; and the held-out test set contained 1,099 images. Consistent random seeds across both experimental notebooks ensure fair partition comparisons. The split strategy is illustrated in Fig. 2 and summarised in Table I.

TABLE I
DATASET SPLIT SUMMARY

Split	Images	Fraction	Purpose
SSL Pre-training Pool	8,789	80.0%	Unsupervised pre-training (labels discarded)
Probe / k-NN Train	1,099	10.0%	Linear probe & k-NN training
Held-out Test	1,099	10.0%	Evaluation only

C. Backbone: DeiT-Small

Both methods use DeiT-Small (`deit_small_patch16_224`) as the backbone, loaded from `timm` with ImageNet pre-trained weights. DeiT-Small partitions 224×224 images into 196 non-overlapping 16×16 patches plus a `[CLS]` token, processing them through 12 transformer blocks with 6 attention heads to produce a 384-dimensional `[CLS]` embedding. In a prior supervised assignment, this backbone achieved 100% accuracy on the same dataset.

D. BYOL Architecture and Training

BYOL consists of an online network (backbone + projector + predictor) and an EMA target network (backbone + projector, no predictor). The projector is a three-layer MLP with BatchNorm (384→4096→4096→256). The predictor is a two-layer MLP (256→4096→256), yielding 43,186,304 total parameters. The EMA momentum τ follows a cosine schedule from 0.996 to 1.0 over 100 epochs. The training objective is the symmetric negative cosine similarity:

$$\mathcal{L}_{\text{BYOL}} = 2 - 2 \cdot \frac{\langle q, z' \rangle}{\|q\| \, \|z'\|} \tag{1}$$

where q is the predictor output and $z' = \text{stop\_grad}(z_{\text{target}})$. Training used AdamW ($lr{=}3{\times}10^{-4}$, $wd{=}1{\times}10^{-6}$) with AMP on 2×Tesla T4 GPUs. Due to wall-clock constraints, training ran for 63 epochs. The BYOL architecture is shown in Fig. 3.

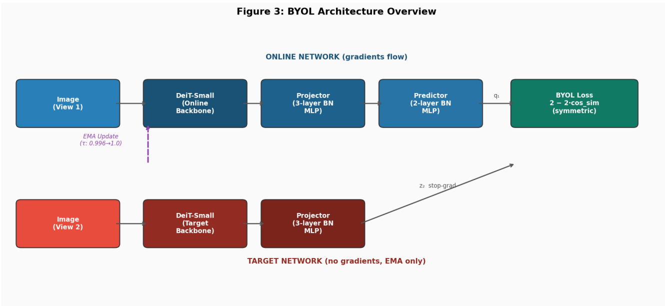

Fig. 3. BYOL architecture. The online network (top) includes backbone, projector, and predictor. The target network (bottom) uses EMA-updated weights with no gradients.

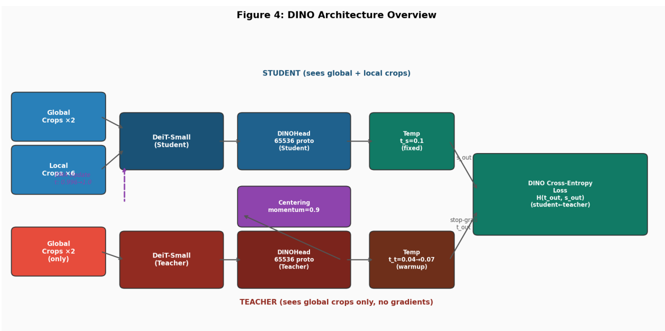

Fig. 4. DINO architecture. The student processes all 8 crops (global + local) while the teacher processes only the 2 global crops, forcing the student to infer global context from local patches.

E. DINO Architecture and Training

DINO employs a student-teacher design with a shared DINOHead architecture:

$$\mathrm{Linear}(384, 2048) \to \mathrm{GELU} \to \mathrm{Linear}(2048, 2048) \to \mathrm{GELU} \to \mathrm{Linear}(2048, 256) \to \ell_2 \to \mathrm{WN}(256, 65536) \quad (2)$$

where WN denotes weight-normalised linear projection. The teacher temperature warms from 0.04 to 0.07 over the first 30 epochs; the student temperature is fixed at 0.1. Running-mean centring (momentum = 0.9) prevents representation collapse. Multi-crop training uses 8 views: 2 global (224×224) and 6 local (96×96). The student processes all 8 views while the teacher processes only the global crops. Training used AdamW (lr=5×10^{-4}, weight decay cosine 0.04→0.4) for 50 epochs. The architecture is shown in Fig. 4.

F. Linear Probing and k-NN Evaluation

After pre-training, the backbone is frozen and a single nn.Linear(384, 5) is trained on the 10% probe set using SGD (lr=0.01, momentum = 0.9) with cosine LR decay over 50 epochs. The k-NN evaluation uses L2-normalised features and cosine similarity for $k \in \{1, 5, 10, 20, 50, 100, 200\}$. Metrics include top-1 accuracy and macro-averaged precision, recall, and F1.

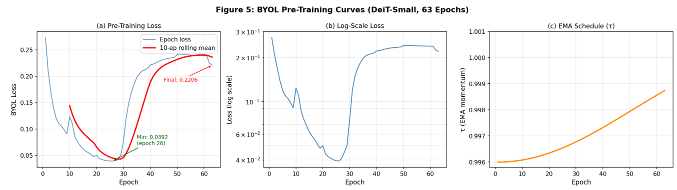

Fig. 5. BYOL pre-training curves: (a) loss with 10-epoch rolling mean, (b) log-scale loss, (c) cosine EMA schedule showing τ rising from 0.996 towards 1.0.

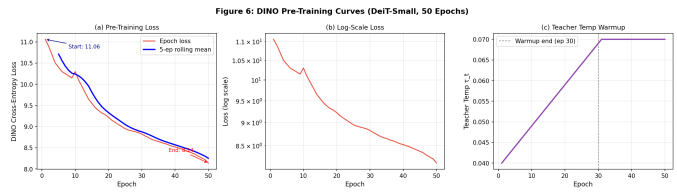

Fig. 6. DINO pre-training curves: (a) loss with rolling mean, (b) log-scale loss, (c) teacher temperature warmup from 0.04 to 0.07 over the first 30 epochs.

IV. Results

A. SSL Pre-Training Convergence

BYOL's symmetric negative cosine similarity loss decreased from 0.2730 to 0.2206 (a 19.2% reduction) over 63 epochs, with a minimum of 0.0392 at epoch 26 before stabilising as the EMA momentum approached 1.0. Fig. 5 shows the training curves.

DINO pre-training (50 epochs) reduced the cross-entropy distillation loss from 11.0613 to 8.1440 (26.4% decrease), with monotonic convergence indicating stable training without collapse. Fig. 6 shows the DINO training dynamics.

B. Linear Probe Results

Table II and Fig. 7 present the linear probing results. DINO outperforms BYOL by approximately 8 pp in top-1 accuracy (88.81% vs. 80.89%) and 7.8 macro F1 points (0.8878 vs. 0.8098). Both methods perform best on powdery mildew (BYOL F1 = 0.9735, DINO F1 = 0.9927) due to its visually distinctive appearance.

C. k-NN Evaluation

Table III and Fig. 8 compare k-NN accuracy across values of k. DINO's k=1 accuracy of 95.27% substantially exceeds its own linear probe accuracy (88.81%), revealing non-linear class boundaries in feature space. BYOL's k-NN performance (72.07% at k=1) falls below its linear probe (80.89%), indicating BYOL features are more linearly separable than geometrically clustered.

D. BYOL Label-Fraction Ablation

Fig. 9 and Table IV show how BYOL representations transfer under different annotation budgets. At just 5% labels, linear probe accuracy reaches 61.33%; at 50% it reaches 76.80%, validating SSL's utility in low-label regimes.

TABLE II
LINEAR PROBE COMPARISON (10% LABELS FOR SSL; 80% LABELS FOR SUPERVISED BASELINE)

Method	Top-1	Prec	Rec	F1
BYOL (DeiT-S, 63 ep)	80.89%	80.99%	81.19%	0.8098
DINO (DeiT-S, 50 ep)	88.81%	89.00%	88.82%	0.8878
Supervised ViT (A01)	100.00%	—	—	—

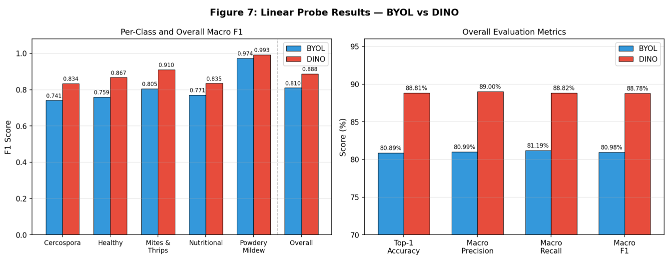

Fig. 7. Linear probe results. Left: per-class and overall macro F1 for BYOL vs. DINO. Right: overall evaluation metrics comparison.

TABLE III
K-NN ACCURACY AT SELECTED k VALUES (COSINE SIMILARITY, L2-NORMALISED FEATURES)

Method	k=1	k=5	k=10	k=20	k=50
BYOL	72.07%	72.07%	71.52%	70.61%	67.79%
DINO	95.27%	88.54%	82.80%	78.80%	74.98%

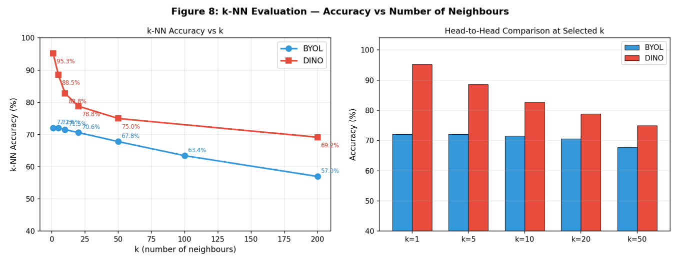

Fig. 8. k-NN accuracy vs. number of neighbours. Left: accuracy curves. Right: head-to-head bar comparison at selected k values.

E. DINO Attention Map Visualisation

Multi-head self-attention from the final DeiT-Small transformer block was extracted via a forward hook. CLS-token attention to all 196 patch positions was averaged across 6 heads and upsampled to 224×224. Diseased regions (fungal lesions in cercospora, white deposits in powdery mildew, stippling in mite infestation) received disproportionately high attention, while background regions received near-zero attention—entirely without label supervision.

V. DISCUSSION

The results demonstrate a clear performance advantage for DINO. Its 88.81% linear probe accuracy exceeds BYOL's 80.89% by approximately 8 pp, achieved with fewer pre-training epochs (50 vs. 63) and only 10% of labels. The k-NN gap is even more pronounced: 95.27% vs. 72.07% at k=1, as shown in Fig. 10.

Several factors explain this gap. DINO's multi-crop strategy (2 global + 6 local views) provides a stronger learning signal than BYOL's two-view augmentation, enabling the model to match local texture patterns with global context—a useful inductive bias for localised leaf lesions. DINO's 65,536-prototype head partitions the feature space far more finely than BYOL's 256-dimensional cosine similarity objective. The DeiT-Small ViT architecture also aligns more naturally with DINO's knowledge distillation objective.

Compared to the supervised Assignment 01 baseline (100% accuracy with 80% labels), a gap of $\sim$11.2 pp for DINO and $\sim$19.1 pp for BYOL remains, despite using only one-eighth of the labelled data. DINO approaches fully supervised performance with dramatically fewer annotations, demonstrating strong data efficiency.

Both runs were wall-clock constrained (BYOL at 63/100 epochs, DINO at 50/100 epochs), so full convergence was not reached. BYOL's label-fraction ablation reveals instability at minimal label counts (1–5%), indicating the need for semi-supervised fine-tuning strategies in extremely low-label regimes.

VI. CONCLUSION AND FUTURE WORK

This paper conducted a comparative analysis of BYOL and DINO self-supervised learning frameworks for chilli disease classification using DeiT-Small. DINO substantially outperformed BYOL, achieving a linear probe accuracy of 88.81% and a k-NN accuracy of 95.27% at k=1, compared to 80.89% and 72.07% for BYOL. DINO's multi-crop strategy and high-dimensional prototypical head produce geometrically richer embeddings. Attention map analysis demonstrates unsupervised disease region localisation with practical implications for precision agriculture.

Future work should: (1) complete 100-epoch training to assess full convergence; (2) broaden comparisons to include SimCLR and MAE; (3) investigate DINOv2 with domain-adapted pre-training data; (4) implement semi-supervised fine-tuning to close the supervised gap; (5) apply DINO attention maps for unsupervised disease segmentation; and (6) evaluate generalisation across additional crop disease datasets.

REFERENCES

[1] J.-B. Grill *et al.*, "Bootstrap Your Own Latent: A New Approach to Self-Supervised Learning," in *Proc. NeurIPS*, 2020.
[2] M. Caron *et al.*, "Emerging Properties in Self-Supervised Vision Transformers," in *Proc. ICCV*, 2021.
[3] M. Oquab *et al.*, "DINOv2: Learning Robust Visual Features without Supervision," *Trans. Mach. Learn. Res.*, 2024.
[4] T. Chen *et al.*, "A Simple Framework for Contrastive Learning of Visual Representations," in *Proc. ICML*, 2020.
[5] K. He *et al.*, "Masked Autoencoders Are Scalable Vision Learners," in *Proc. CVPR*, 2022.
[6] A. Dosovitskiy *et al.*, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," in *Proc. ICLR*, 2021.
[7] H. Tran *et al.*, "Chilli Disease Detection Dataset," *Mendeley Data*, v2, 2024. doi: 10.17632/tf9dtfz9m6.2.

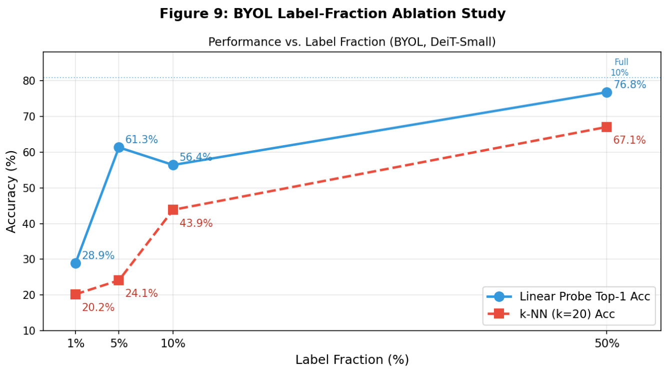

Fig. 9. BYOL label-fraction ablation study. Linear probe (blue) and k-NN k=20 (red) accuracy as a function of available label fraction.

TABLE IV
BYOL LABEL-FRACTION ABLATION STUDY

Label Fraction	**N Labels**	**LP Top-1**	**k-NN k=20**
1%	$\sim$11	28.94%	20.20%
5%	$\sim$55	61.33%	24.11%
10%	$\sim$110	56.41%	43.86%
50%	$\sim$550	76.80%	67.06%

[8] I. Misra and L. van der Maaten, “Self-Supervised Learning of Pretext-Invariant Representations,” in *Proc. CVPR*, 2020.

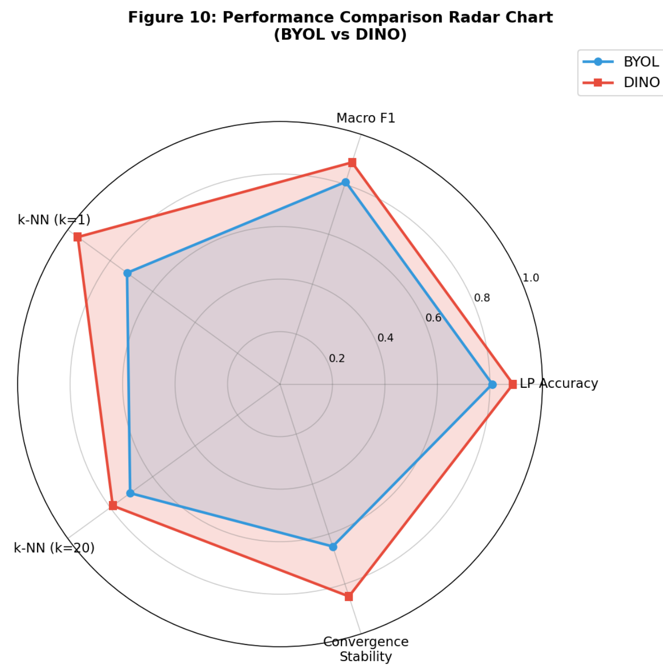

Fig. 10. Radar chart comparing BYOL and DINO across five performance dimensions. DINO consistently dominates, especially in k-NN accuracy metrics.